



Constellation

James A. FitzPatrick NPP
P.O. Box 110
Lycoming, NY 13093

Timothy C. Peter
Site Vice President- JAF

JAFP-23-0017
April 19, 2023

United States Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, D.C. 20555-0001

James A. FitzPatrick Nuclear Power Plant
Renewed Facility Operating License No. DPR-059
NRC Docket No. 50-333

Subject: LER: 2023-001-00, Primary Containment Isolation System Isolation due to Initiation of Main Condenser Fire Protection Foam System

Dear Sir or Madam:

This report is being submitted pursuant to 10 CFR 50.73(a)(2)(iv)(A).

There are no new regulatory commitments contained in this report.

Questions concerning this report may be addressed to Mr. Mark Hawes, Regulatory Assurance Manager (Acting), at (315) 349-6659.

Sincerely,

Timothy C. Peter
Site Vice President

TCP/MH/pa

Enclosure: LER: 2023-001-00, Primary Containment Isolation System Isolation due to Initiation of Main Condenser Fire Protection Foam System

cc: USNRC, Region I Administrator
USNRC, Project Manager
USNRC, Resident Inspector
INPO Records Center (IRIS)

NRC FORM 366 (03-14-2023)		U.S. NUCLEAR REGULATORY COMMISSION			APPROVED BY OMB: NO. 3150-0104 EXPIRES: 08/31/2023							
		LICENSEE EVENT REPORT (LER) (See Page 3 for required number of digits/characters for each block)										
		(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)										
1. Facility Name James A. FitzPatrick Nuclear Power Plant					☒ 050 ☐ 052		2. Docket Number 05000333		3. Page 1 OF 3			
4. Title Primary Containment Isolation System Isolation due to Initiation of Main Condenser Fire Protection Foam System												
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved			
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050 Docket Number		
02	19	2023	2023 - 001 - 00			04	20	2023	N/A	N/A		
9. Operating Mode 1					10. Power Level 100							
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)												
10 CFR Part 20		☐ 20.2203(a)(2)(vi)		10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		☐ 73.1200(a)		
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)		☐ 73.1200(b)		
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)		☐ 73.1200(c)		
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.36(c)(2)		☒ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)		☐ 73.1200(d)		
☐ 20.2203(a)(2)(i)		10 CFR Part 21		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(A)		10 CFR Part 73		☐ 73.1200(e)		
☐ 20.2203(a)(2)(ii)		☐ 21.2(c)		☐ 50.69(g)		☐ 50.73(a)(2)(v)(B)		☐ 73.77(a)(1)		☐ 73.1200(f)		
☐ 20.2203(a)(2)(iii)				☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(2)(i)		☐ 73.1200(g)		
☐ 20.2203(a)(2)(iv)				☐ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(ii)		☐ 73.1200(h)		
☐ 20.2203(a)(2)(v)				☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(vii)						
☐ OTHER (Specify in Abstract below or in NRC Form 366A).												
12. Licensee Contact for this LER												
Licensee Contact Mr. Mark Hawes, Regulatory Assurance Manager (Acting)									Telephone Number (Include Area Code) 315-349-6659			
13. Complete One Line for each Component Failure Described in this Report												
Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS			
14. Supplemental Report Expected								15. Expected Submission Date		Month	Day	Year
☒ No		☐ Yes (If yes, complete 15. Expected Submission date)										
Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)												
At 0105 EST on February 19, 2023, with the James A. FitzPatrick Nuclear Power Plant (JAF) at 100 percent power, a High Main Steam Line Radiation signal initiated a primary containment isolation valve closure signal in multiple systems. During the performance of FPP-3.25 "Fire Protection Manifold #5 Sprinkler System Main Drain and Water Flow Alarm Test", the main condenser foam system was inadvertently initiated, resulting in firefighting foam intrusion into a sump which migrated to the main condenser and eventually to the reactor coolant system. The cause was determined to be a human performance related error during testing. Corrective actions include conducting human performance training and evaluations and hardening of the impacted plant sump system.												

NRC FORM 366A (03-14-2023)		U.S. NUCLEAR REGULATORY COMMISSION		APPROVED BY OMB: NO. 3150-0104		EXPIRES: 08/31/2023	
		LICENSEE EVENT REPORT (LER) CONTINUATION SHEET					
(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)				Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollects.Resource@nrc.gov , and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: oira_submission@omb.eop.gov . The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.			
1. FACILITY NAME James A. FitzPatrick Nuclear Power Plant		☒ 050 ☐ 052	2. DOCKET NUMBER 05000333		3. LER NUMBER		
					YEAR 2023	SEQUENTIAL NUMBER – 001	REV NO. – 00

NARRATIVE

Background

James A. FitzPatrick Nuclear Power Plant (JAF) has two firefighting foam systems: the main condenser foam system [EIS identifier: KQ], which is the subject of this event, and the High Pressure Coolant Injection (HPCI) foam System. The purpose of main condenser foam system is to provide fire extinguishment for the Main Condenser Liquid Holding Pit. The foam system was provided on the basis that the oil from any major turbine oil leak would drain to the pit and a foam system would be effective in fire extinguishment in a confined pit area.

Inside the same building, the turbine building equipment drain sump [WK], also known as the low conductivity sump, is designed to collect, and hold leakage from turbine building equipment drains. During power operation, the low conductivity water in the sump is drawn into the main condenser [SD] with vacuum drag. This water is combined with main condenser condensate water and returned to the Reactor Coolant System (RCS) [AB]. Two pumps are also available to move water out of the sump to the condenser or radwaste treatment.

Event Description

On February 18, 2023, at 23:00, with the JAF at 100 percent power, a scheduled test began for FPP-3.25, Fire Protection Manifold #5 Sprinkler System Main Drain and Water Flow Alarm Test. The purpose of the test is to demonstrate water supply is available to the sprinkler system supplied from Manifold #5 and to demonstrate operability of the waterflow alarm. The step in the procedure being performed by the control room stated to "Press Acknowledge button at Panel FPP and verify Red CONDENSER PIT FOAM SYS light is on, and alarm bell is off"; however, the operator depressed the main condenser foam system initiation pushbutton instead of the acknowledge pushbutton, at 00:42 on February 19.

The foam system injected, per design, into the Main Condenser Liquid Holding Pit and a foam/water mixture migrated to the nearby turbine building equipment drain sump. The main condenser vacuum drag drew the foam/water mixture out of the sump into the main condenser hotwell which was then introduced into the RCS via the condensate and feedwater system [JB].

At 01:06, Main Steam Line Hi-Hi radiation actuated Primary Containment Isolation System (PCIS) [JM] Group 1 isolation signal. Reactor Water Recirculation (RWR) [AD] sample valve 02-2AOV-39 and 02-2AOV-40 closed. The isolation signal also isolated the normally closed Main Steam Line Drain System [SB] and Condenser Air Removal System [SH].

At 01:14, Operators isolated the high conductivity sump vacuum drag flow path and aligned it to radwaste. This stopped the intrusion of high conductivity foam into the condenser. Conductivity, off-gas radiation, and main steam line radiation then returned to pre-transient values.

This event was reported in accordance with 10 CFR 50.72(b)(3)(iv)(A) via ENS 56373. This report is submitted in accordance 10 CFR 50.73(a)(2)(iv)(A).

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Event Analysis

The initiation of the main condenser foam system resulted in high conductivity water/foam mixture accumulating in the Main Condenser Liquid Holding Pit flooded the turbine building equipment drain sump. Through the vacuum drag process, the sump contents were drawn into the main condenser, mixed with condensate water, and then introduced into the reactor coolant.

Fire foam contains organic compounds which breakdown in the presence of radiation in the reactor. One by-product, Oxygen-16, becomes activated by a neutron and decays into the isotope Nitrogen-16. Nitrogen-16 has a very short half-life of 7.13 seconds. When it decays, the radiation was detected by the Main Steam radiation monitors and resulted in the Main Steam Line Hi-Hi radiation signal. The signal is designed to detect potential fuel failures or a postulated Control Rod Drop Accident. However, the increase in Nitrogen-16 was much larger than the background level designed for during normal full power operation. The radiation increase during this event was temporary and returned to normal levels when the turbine building equipment drain sump was isolated from the main condenser.

Cause

The cause of this event was a failure of the Operator to self-check when performing the task resulting in a human performance error and the wrong pushbutton being depressed.

Similar Events

No similar events were identified at JAF.

Corrective Actions

Completed Actions

Physical Plant: Sealed the openings on the top of the turbine building equipment drain sump and installed a berm around the sump.

Barrier Control: Placed signage on fire protection panel door requiring permission to open the door.

Performance Management: Qualifications were removed from the operator and remediation along with further evaluation completed.

Planned Actions

Perform a human performance evaluation with Reactor Operators and Equipment Operators. Determine training needs based on the results of the evaluation.

Safety Significance

Nuclear safety – **There were no actual radiological consequences.** The Nitrogen-16, produced as a result of the water/fire foam mix being introduced reactor coolant system, decayed quickly, and has a short half-life of just over 7 seconds. The Offgas System [WF] is designed to hold up isotopes in piping where it has time to decay to very low levels.

References

Issue Report - IR 04555544, Inadvertent Initiation of Condenser Foam, dated February 19, 2023